

Consumer welfare beyond GDP

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Abstract

I construct individual and collective welfare when consumers have heterogeneous preferences, unequal incomes, and face different prices. The theory builds on three axioms: collective welfare (i) is maximized at *Laissez-faire* when consumers have identical budgets, (ii) treats same-preference consumers anonymously, (iii) is separable across economies. I prove a new incompatibility between them whenever there is price dispersion. Then, possibilities are obtained when separability only applies across proportional economies. As standard, these possibilities include a sum of expenditure functions evaluated at a common price. The main novelty is that the latter is an income-weighted harmonic mean of individual prices. I apply these results to the measurement of costs and standards of living. For international comparisons of real GDP, one can modify PPP indices to make them robust to preference heterogeneity without additional data. For growth and inflation measurement, one must adopt constant base prices reflecting all available years.

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1 Introduction

Economists routinely compare heterogeneous consumers. For example, because countries with different income levels typically also differ in prices, we deflate expenditures by purchasing power parities (PPP) indices to retrieve a comparable measure of *real* income. While Neary (2004) showed that the Penn World Table's PPP can be rationalized by a representative agent, Deaton and Heston (2010) stressed that this practice neglects preference heterogeneity across countries, leaving the welfare foundations of these comparisons unclear.

The present paper proposes a theory of individual welfare when consumers have heterogeneous preferences, unequal income, and face different prices. The theory relies on axioms that pin down collective consumer welfare. Because social welfare functions entail interpersonal comparisons of individuals, this measure of collective consumer welfare implies the relevant measure of individual consumer welfare. In turn, the comparison of individuals who face different prices informs on how to deflate expenditures.

The theory uses three kinds of axioms. The first axiom imposes that, if all agents had the same budgets, letting them choose their preferred bundle in that budget, i.e. *Laissez-faire*, yields an allocation better than any other feasible allocations. The second axiom requires that if two agents have identical preferences but unequal budgets, swapping their consumption bundles does not affect collective welfare. The third axiom is the standard Fleming (1952) Separability¹ whereby indifferent agents do not matter, i.e. changes in their budgets, preferences, or consumption do not affect collective welfare as long as they remain indifferent.

These axioms are compatible with any degree of aversion to inequality. They nest the case of equality of opportunity, where the observer is averse to consumption inequalities due to unequal budget but not averse to consumption inequalities due to heterogeneous preferences. As such, the approach resonates with the debate on building economic indicators *beyond GDP* (Fleurbaey, 2009; Fleurbaey & Blanchet, 2013).

The first main result shows that these axioms are incompatible when there is price dispersion. To see why, consider two different economies where all agents have identical budgets. Because both prices and income in the first economy may be different from the ones in the second economy, it may be that some allocations are feasible in both of them. Then, we can construct cases where the *Laissez-faire* allocation of the first economy is feasible in the second, and *vice versa*. This is enough to engender impossibilities: the first axiom ranks the *Laissez-faire* allocation as optimal in the first economy and the other axioms imply a stability of that ranking between the two economies, but the first axiom ranks another allocation as optimal in the second economy.

The paper argues that one should escape the impossibility by weakening the third axiom.

¹Fleming (1952) proved that this Separability axiom, combined with Strong Pareto, leads to a utilitarian social welfare function

Intuitively, changes in indifferent agents' budgets are not innocuous because they directly affect what is feasible in the economy, which matters through the first axiom. I propose to apply Separability between two economies only if they are proportional to one another, i.e. when the total resources in good l in the first economy is multiplied by $\lambda > 0$ in the second economy, for all goods. This is enough to escape the impossibility. I prove several representation theorems of social welfare functions that can be used to compare the welfare of consumers interpersonally and collectively.

The second main result of this paper is that the individual consumer welfare measure should be an expenditure function that uses an identical price vector for all consumers. This common price vector reflects the purchasing power of the economy: the price of good l is the ratio between the economy's aggregate income and the total quantity of good l in resources. Equivalently, the common price for a good is the income-weighted harmonic mean of individual prices.

The third set of results aggregates this individual consumer welfare into collective consumer welfare. I use known axiomatic results and show that the aggregator can be a sum or a maximin operator. Interestingly, the degree of inequality aversion in the social welfare function follows a one-to-one relationship with the observer's inequality aversion between consumers with identical prices. This suggests that welfare must be constructed in a two-step procedure: first, compare interpersonally heterogeneous consumers using their expenditure function with the common price; second, apply the same inequality aversion between them as you would if there was no price dispersion.

Because no restrictions are imposed on the structure of preference nor on the extent of income and price dispersion, the results provide welfare foundations for a large set of cost and standards of living measurement problems. The last part of the paper proposes several empirical applications.

For international comparisons of real GDP, the paper recommends adopting a different Purchasing Power Parity (PPP) index than what is typically done. While the standard Geary-Khamis international dollar is based on a consumption-weighted harmonic mean of countries' prices, this paper argues for substituting consumption weights with income weights. Intuitively, consumption weights are ill-suited for international comparisons because, contrary to income weights, they are not invariant to preference shocks.

This welfare-consistent PPP index has at least three interesting properties. First, and maybe most importantly, it does not require more data than what is collected by statistical agencies.² Second, it is orthogonal to the planner's inequality aversion. Third, it is plutocratic, i.e. influenced more by larger countries, without conflicting with fairness views. Overall, the inquiry suggests that most of Deaton and Heston (2010)'s concerns can be alleviated at little empirical costs.

²This has consequences not only for real GDP but also for its derived statistics to measure labor productivity, poverty, or inequality as in the Distributional National Accounts as in Piketty, Saez, and Zucman (2018).

Finally, I also show the implications of the paper for the problem of measuring inflation and growth in one country over time. In that context, the paper provides a foundation for constructing a base year that is consistent with welfare.

Literature

Comparing agents with heterogeneous preferences is notoriously difficult since Arrow (1950)'s impossibility theorem. Indeed, there is no obvious way of summing up utilities when utility functions differ across individuals. The present paper escapes Arrow's impossibility as does the social choice theory of fairness initiated by Fleurbaey and Maniquet (2006, 2011, 2018). It uses information on indifference curves, while Arrow's theorem only used utility levels. This allows us to characterize welfare measures in the realm of *equivalent income*, such as willingness to pay, expenditure functions, or money-metric utility functions. For example, the Hicksian compensating and equivalent variations belong to this class.

A lasting³ critique of these expenditure functions is that they depend on a reference price that is left at the researcher's discretion (see discussions in e.g. Bosmans, Decancq, and Ooghe (2018), Capéau, Decoster, and De Sadeleer (2023), and Fleurbaey and Blanchet (2013)). This limitation motivated the study of reference-independent welfare assessment (Blackorby, Laisney, & Schmachtenberg, 1993; Eden & Freitas, 2024; Roberts, 1980). The main contribution of this paper with respect to the social choice literature is to axiomatize welfare measures based on expenditure functions whose reference price is endogenized by the axioms. This comes from the fact that I consider price dispersion across individuals, while previous papers imposed homogeneous prices for all agents, building on the Walrasian tradition in general equilibrium theory. Yet, the law of one price does not hold in many datasets.⁴ This additional dimension of heterogeneity engenders an impossibility, which in turn leads to new characterization results.

This paper also contributes to the index number theory as initiated by Fisher (1922). The index number problem consists in building price and quantity indices based on dataset of observed prices and expenditures for each consumer. van Veelen and van der Weide (2008) divide the proposed solutions in two approaches. The axiomatic approach imposes series of desirable mathematical properties on the indices but is typically not linked with classical demand theory (Samuelson & Swamy, 1974; Van Veelen, 2002). The economic approach rationalizes observed differences from a representative consumer demand system (Diewert, 1976; Neary, 2004). The present paper rejects a representative consumer unlike the latter but has links with classical demand theory unlike the former. Moreover, the paper

³A classical critique by Blackorby and Donaldson (1988) was that money-metric utility function may lead to inequalitarian assessment. However, Fleurbaey and Maniquet (2011) and Schlee and Khan (2022, 2023) showed multiple ways to have inequality-averse social welfare functions based on money-metric utility.

⁴For example, the Penn effect describes the fact that non-tradable goods are more expensive in richer countries, as documented by Kravis, Heston, and Summers (1978) and explained in trade theory by Balassa (1964) and Samuelson (1964) (see Samuelson (1994) for a review). An early observation of the failure of the law of one price is in Stigler (1961).

does not impose any domain restrictions on individual preferences, such that its results are valid whether preferences are homothetic (Diewert & Nakamura, 1993) or not (Jaravel & Lashkari, 2024).

The paper is structured as follows. In section 2, I present the standard consumption space and define the objects of interest. In section 3, I present the key axioms and prove their incompatibilities. In section 4, I weaken separability and derive several representation theorems. In section 5, I derive implications for various applications. In section 6, I conclude.

2 Model

The environment is a standard consumption space. A society $\mathcal{N} = \{1, \dots, N\}$ is composed of a finite number $N \geq 4$ of consumers. There are $L \geq 2$ private goods and a consumer i 's consumption bundle is denoted by $z_i \in \mathbb{R}_+^L$.

Agents are heterogeneous in their preference ordering $\succsim_i$ which is assumed to be complete, transitive, continuous, locally non-satiated, and convex, as well as in their endowments in goods⁵ $\omega_i \in \mathbb{R}_+^L$ and nonzero personalized prices $p_i \in \mathbb{R}_{++}^L$. Personalized prices may include the standard case of no price dispersion in a Walrasian equilibrium, as well as price dispersion due, for example, to search frictions, transport costs, price discrimination, regulations, heterogeneous needs, or comparative advantages in production. Because the paper remains agnostic about the origins of prices, its results apply regardless of how prices are formed. For brevity, I will refer to the scalar product $p_i \cdot \omega_i$ as income and denote it by $y_i = p_i \cdot \omega_i \geq 0$.

An allocation z is a list of consumption bundles for each agent $z = (z_1, z_2, \dots, z_N) \in \mathbb{R}_+^{LN}$ and it may or may not be rationalizable by agents' utility maximization constrained by a budget set. An economy e is composed of N agents which are heterogeneous in preferences, endowments, and prices $e = \{(\succsim_i, \omega_i, p_i)_{i \in \mathcal{N}}\}$. The set of all economies is denoted by $\mathcal{E}$. I assume full domain, except that $\mathcal{E}$ does not contain the uninteresting case where all agents have no endowments. Hence, there exists at least one $i \in \mathcal{N}$ with $\omega_i > 0$.

For each economy e , the observer has a complete and transitive ordering $\mathbf{R}(e)$ over allocations $z \in \mathbb{R}_+^{LN}$. This will be the main object of interest of the paper. I use $\mathbf{R}(e)$, $\mathbf{P}(e)$, $\mathbf{I}(e)$ for weak preference, strict preference, and indifference, respectively.⁶

Because social welfare is constructed in the first-best, the observer can use lump-sum transfers. It will be useful to define the following two concepts. An allocation $\hat{z}$ is feasible by lump-sum transfers from the economy e if there exists a vector of individual lump-sum

⁵Observe that this model nests a model where one good is set as the numéraire and endowments are only in that good, that is a monetary endowment. All our results are independent of the choice of the numéraire.

⁶The paper builds welfare at an allocation conditional on primitive preferences, incomes, and prices. It does so for any primitive and any allocation. Hence, the positive question of how a shock to primitives changes the equilibrium allocation does not affect the paper's theorems.

transfers $(t_1, t_2, \dots, t_N) \in \mathbb{R}^N$ with $\sum_{i \in \mathcal{N}} t_i \leq 0$ such that bundles $\hat{z}_i$ are part of individuals' budgets $B(t_i, \omega_i, p_i)$:

$$\hat{z}_i \in B(t_i, \omega_i, p_i) = \{z_i \in \mathbb{R}_+^L : p_i \cdot z_i \leq p_i \cdot \omega_i + t_i\}$$

These allocations $\hat{z}$ are objects in $\mathbb{R}_+^{LN}$. For geometric intuitions below, it will be useful to define the following concept in $\mathbb{R}_+^L$. The aggregate budget of an economy, denoted by $\mathcal{B}(e)$, describes the economy's raw resources: it is the set that collects all the consumption vectors that can be consumed before any lump-sum transfers. Formally, it is obtained as the (Minkowski) sum of individual budgets⁷, i.e.

$$\mathcal{B}(e) = \left\{ \sum_{i \in \mathcal{N}} z_i \in \mathbb{R}_+^L : z_i \in B(0, \omega_i, p_i) \quad \forall i \in \mathcal{N} \right\}$$

The logical relationship between these two concepts is summarized in the next remark.

Remark 1. *In equal-price economies $\{(\omega_i, p_i)_{i \in \mathcal{N}}\}$, an allocation is feasible by lump-sum transfers $\hat{z}$ if and only if $\sum_i \hat{z}_i \in \mathcal{B}(e)$.*

This equivalence only holds in economies without price dispersion. Indeed, with price dispersion, lump-sum transfers from individuals with high prices to individuals with low prices expand consumption possibilities.

3 Impossibilities

The axiomatic construction of the social ordering $\mathbf{R}(e)$ consists of imposing desirable properties, i.e., axioms, which restrict the scope of possibilities among the universe of transitive and complete $\mathbf{R}(e)$. This section introduces the key axioms of the paper and shows that they lead to an impossibility.

Before introducing the three key axioms of this paper's theory, let me briefly state the two standard versions of the Pareto principle, which will be useful later in the paper.

Weak Pareto: $\forall e \in \mathcal{E}$ and all allocations z, z' , if $z_i \succ_i z'_i \quad \forall i \in N$, then $z \mathbf{P}(e) z'$.

Strong Pareto: $\forall e \in \mathcal{E}$ and all allocations z, z' , if $z_i \succeq_i z'_i \quad \forall i \in N$, then $z \mathbf{R}(e) z'$. If, in addition, $z_j \succ_j z'_j$ for some $j \in N$, then $z \mathbf{P}(e) z'$.

The first main axiom, **Laissez-faire**, embodies the view that agents deserve the consequences of their tastes.⁸ Formally, it requires that when there is no cross-sectional heterogeneity in prices and endowments, the Laissez-faire allocation, where each agent freely

⁷While Minkowski summation preserves convexity, it does not necessarily preserve linearity. In general, $\mathcal{B}(e)$ will be a piecewise-linear convex set and it coincides with the simplex if and only if $p_i = p_j$ for all $i, j \in \mathcal{N}$.

⁸Political philosophers such as Arneson (1990), Cohen (1989), Dworkin (1981), and Rawls (1971) all defended variants of this principle, as discussed in Fleurbaey (2008) and Fleurbaey and Maniquet (2006, 2011).

chooses what they prefer out of their budget set $B(0, \omega, p)$, is strictly preferred to any other allocations feasible by lump-sum transfers from the economy.

Laissez-faire: $\forall e \in \mathcal{E}$ such that $(\omega_i, p_i) = (\omega, p)$ for all $i \in \mathcal{N}$, we have

$$z^* \mathbf{P}(\mathbf{e}) \hat{z}$$

for z^* such that $z_i^* \in \max_{\succsim_i} B(0, \omega, p)$ for all $i \in \mathcal{N}$ and any $\hat{z} \neq z^*$ feasible by lump-sum transfer for e .

This axiom deserves three comments. First, remember that our setting is agnostic about the origins of prices. In the particular case where prices are formed by general equilibrium trades of equally-endowed agents, z^* is the Walrasian equilibrium. Hence, this axiom imposes a preference for the competitive equilibrium in that particular case, but it also bites beyond that: the Laissez-faire allocation is preferred to any other feasible allocation whenever there is no price dispersion across agents with equal incomes, for example, because of regulations.

Second, in the context of consumer welfare measurement, this axiom implies that, when everyone has the same budget, welfare should be constructed based on an agent's own favorite bundle rather than any other bundle. For example, this axiom precludes the economist from choosing an arbitrary bundle and evaluating distances to it as welfare, such as a pre-determined poverty bundle or a given CPI consumer basket.

Third, in and of itself, this axiom does not force the planner to be inequality averse. However, for an inequality-averse observer, it imposes that the part of consumption inequalities that are due to heterogeneous tastes are unproblematic. This means that interpersonal inequalities in welfare may not be driven by heterogeneous preferences. Relatedly, a resource allocation which fully equalizes consumption is not optimal if this axiom is imposed because part of consumption inequalities should not be reduced as they reflect unequal tastes.

The second axiom, **Same-Preference Anonymity**, reflects the view that agents do not deserve the consequences of their budgets. In particular, the axiom imposes that when a pair of agents have identical preferences, a permutation of their bundles should be socially indifferent.⁹ In other words, the social ranking is anonymous among agents with identical preferences.

Same-Preference Anonymity: $\forall e \in \mathcal{E}$ where agents i, j are such that $\succsim_i = \succsim_j$ then

$$(z_i, z_j, z_{\mathcal{N} \setminus \{i, j\}}) \mathbf{I}(\mathbf{e}) (z_j, z_i, z_{\mathcal{N} \setminus \{i, j\}}).$$

This axiom introduces a mild degree of compensation in the social welfare function. Indeed, **Same-Preference Anonymity** says that swapping the bundles from agents with large

⁹This type of axioms also received support in political philosophy among liberal egalitarians (Dworkin, 1981). It is reminiscent to Suppes (1966)'s grading principle. See Hammond (1976) for an early treatment and Maniquet (2004) for its relationship to equality of opportunity.

budgets to agents with small budgets does not change social welfare, but neither does the opposite. As a consequence, agents do not deserve the consequences of their budgets. Equivalently, welfare should be constructed in the same way for all agents with the same preferences. In the context of measuring consumer welfare across countries, this axiom forbids comparing equal-preference countries with country-specific measuring rods.

Taken together, **Laissez-faire** and **Same-Preference Anonymity** do not determine how much inequality should society dislike, but they rather restrict the information that can be used when constructing welfare. If these two axioms were combined with an inequality-aversion requirement, they would lead to a social welfare function that reflects the ethics of equality of opportunity: consumption inequalities arising from unequal budgets should be reduced, but consumption inequalities arising from unequal preferences should not.

Observe that both axioms only apply under some specific premises, i.e. for some but not all e in $\mathcal{E}$. In order to obtain a complete ordering $\mathbf{R}(\mathbf{e})$ for all economies, one must use a cross-economy robustness condition. Since the characterization of utilitarianism by Fleming (1952), the standard cross-economy condition requires that indifferent agents should not have a say in the social evaluation between two alternatives. This is sometimes defended on democratic grounds: only those effectively affected by the alternatives should be heard. More formally, **Fleming (1952) Separability** requires that if some agents $\mathcal{K}$ are receiving the same bundle in two allocations, then the social ordering between these two allocations is unaffected if these agents change their preferences, endowments, prices, or receive a different bundle.

Fleming (1952) Separability: $\forall e, e' \in \mathcal{E}$ and the partitions $\mathcal{K} + \mathcal{M} = \mathcal{N}$ with $e_{\mathcal{K}} = \{(\succsim_i, \omega_i, p_i)_{i \in \mathcal{K}}\}$, $e'_{\mathcal{K}} = \{(\succsim'_i, \omega'_i, p'_i)_{i \in \mathcal{K}}\}$, $e_{\mathcal{M}} = \{(\succsim_i, \omega_i, p_i)_{i \in \mathcal{M}}\} = e'_{\mathcal{M}}$, and z, z' then,

$$(z_{\mathcal{K}}, z_{\mathcal{M}})\mathbf{R}(e_{\mathcal{K}}, e_{\mathcal{M}})(z'_{\mathcal{K}}, z'_{\mathcal{M}}) \iff (z'_{\mathcal{K}}, z_{\mathcal{M}})\mathbf{R}(e'_{\mathcal{K}}, e_{\mathcal{M}})(z'_{\mathcal{K}}, z'_{\mathcal{M}})$$

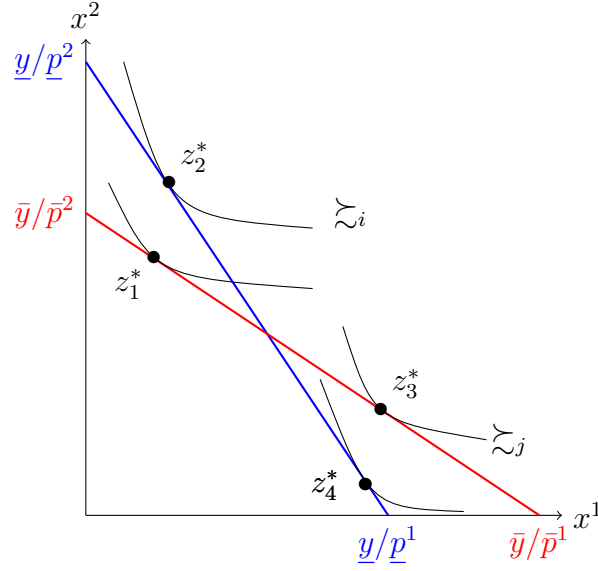
I note that the version presented here is logically weaker than the axioms of separability in subpopulations used in d'Aspremont and Gevers (1977), Fleurbaey (2003), and Maniquet (2004). The next theorem shows that the three axioms outlined above lead to an impossibility when combined together.

Theorem 1. *There does not exist any $\mathbf{R}(\mathbf{e})$ that satisfies **Laissez-faire**, **Same-Preference Anonymity**, and **Fleming (1952) Separability**.*

Proof. By contradiction, assume that the statement does not hold. We will work with $N = 4$ economies constructed from two preference profiles $\succsim_i$ and $\succsim_j$ and two Laissez-faire budgets $B(0, \bar{\omega}, \bar{p})$ and $B(0, \underline{\omega}, \underline{p})$. There are four bundles of interest that maximize either $\succsim_i$ or $\succsim_j$ over the budgets $B(0, \bar{\omega}, \bar{p})$ and $B(0, \underline{\omega}, \underline{p})$ and we denote them by $z_1^*, z_2^*, z_3^*, z_4^*$ respectively. Moreover, preferences and budgets are chosen such that $z_1^*, z_2^*, z_3^*, z_4^*$ are the vertices of a parallelogram. The situation, whose existence is guaranteed in $\mathcal{E}$, is depicted in Figure 1.

Geometrically, the diagonals of a parallelogram bisect each other, such that $\frac{z_1^* + z_3^*}{2} = \frac{z_2^* + z_4^*}{2}$. This implies that both Laissez-faire allocations use an identical amount of aggregate

Figure 1: Illustration of the proof of Theorem 1 where $\bar{y} = \bar{p}\bar{\omega}$ and $\underline{y} = \underline{p}\underline{\omega}$.



Notes: The quantity of good 1 is denoted by x_1 on the horizontal axis and the quantity of good 2 is denoted by x_2 on the vertical axis. The economies have been chosen such that the Laissez-faire allocation in the blue economy is feasible by lump sum transfer from the red economy, and vice versa, that is $\frac{z_1^* + z_3^*}{2} = \frac{z_2^* + z_4^*}{2}$.

resources

$$z_1^* + z_1^* + z_3^* + z_3^* = z_2^* + z_2^* + z_4^* + z_4^*$$

Hence, by Remark 1, $(z_1^*, z_1^*, z_3^*, z_3^*)$ is feasible by lump-sum transfers from economy where all agents have the blue budget, and $(z_2^*, z_2^*, z_4^*, z_4^*)$ is feasible by lump-sum transfers from the economy where all agents have the red budget.

By **Laissez-faire**,

$$(z_1^*, z_1^*, z_3^*, z_3^*) \quad \mathbf{P} \left(\left\{ (\bar{\omega}, \bar{p}), (\bar{\omega}, \bar{p}), (\bar{\omega}, \bar{p}), (\bar{\omega}, \bar{p}) \right\} \right) \quad (z_2^*, z_2^*, z_4^*, z_4^*)$$

By **Fleming (1952) Separability**,

$$(z_1^*, z_2^*, z_4^*, z_3^*) \quad \mathbf{P} \left(\left\{ (\bar{\omega}, \bar{p}), (\underline{\omega}, \underline{p}), (\underline{\omega}, \underline{p}), (\bar{\omega}, \bar{p}) \right\} \right) \quad (z_2^*, z_2^*, z_4^*, z_4^*)$$

By **Same-Preference Anonymity**,

$$(z_2^*, z_1^*, z_3^*, z_4^*) \quad \mathbf{I} \left(\left\{ (\bar{\omega}, \bar{p}), (\underline{\omega}, \underline{p}), (\underline{\omega}, \underline{p}), (\bar{\omega}, \bar{p}) \right\} \right) \quad (z_1^*, z_2^*, z_4^*, z_3^*)$$

By **Transitivity**,

$$(z_2^*, z_1^*, z_3^*, z_4^*) \quad \mathbf{P} \left(\left\{ (\bar{\omega}, \bar{p}), (\underline{\omega}, \underline{p}), (\underline{\omega}, \underline{p}), (\bar{\omega}, \bar{p}) \right\} \right) \quad (z_2^*, z_2^*, z_4^*, z_4^*)$$

By **Fleming (1952) Separability**,

$$(z_2^*, z_1^*, z_3^*, z_4^*) \quad \mathbf{P} \left(\left\{ (\zeta_i, \underline{\omega}, \underline{p}), (\zeta_i, \underline{\omega}, \underline{p}), (\zeta_j, \underline{\omega}, \underline{p}), (\zeta_j, \underline{\omega}, \underline{p}) \right\} \right) \quad (z_2^*, z_2^*, z_4^*, z_4^*)$$

But this contradicts **Laissez-faire**, which completes the proof. ■

A natural attempt to escape the impossibility would consist in weakening **Laissez-faire** to a weak social preference for the Laissez-faire allocation rather than a strict preference. This is what the next axiom, **Laissez-faire^W**, achieves.

Laissez-faire^W: $\forall e \in \mathcal{E}$ such that $(\omega_i, p_i) = (\omega, p)$ for all $i \in \mathcal{N}$, we have $z^* \mathbf{R}(e) \hat{z}$ for z^* such that $z_i^* \in \max_{\zeta_i} B(0, \omega, p)$ for all $i \in \mathcal{N}$ and any $\hat{z} \neq z^*$ feasible by lump-sum transfer for e .

The next theorem shows that the impossibility persists with **Laissez-faire^W** if one looks for orderings that satisfy the widely-used **Weak Pareto** principle. The proof builds on the same intuitions as Theorem 1 and it is relegated to Appendix A.

Theorem 2. *There does not exist any $\mathbf{R}(e)$ that satisfies **Weak Pareto**, **Laissez-Faire^W**, **Same-Preference Anonymity**, and **Fleming (1952) Separability**.*

These impossibilities are somewhat surprising¹⁰ because Bosmans, Decancq, and Ooghe (2018), Fleurbaey and Maniquet (2011, 2017), and Piacquadio (2017) all characterized possibilities with similar axioms.¹¹ The present paper only differs by allowing price dispersion in the cross-section and this is enough to create impossibilities. Indeed, price dispersion implies that budgets may be non-nested so the optimal Laissez-faire allocation in one economy is feasible in the other economy, and *vice versa*. Because **Laissez-faire** imposes a different optimal allocation in both economies but **Fleming (1952) Separability** requires a stability of the social ordering across economies, we reached impossibilities.

4 Possibilities

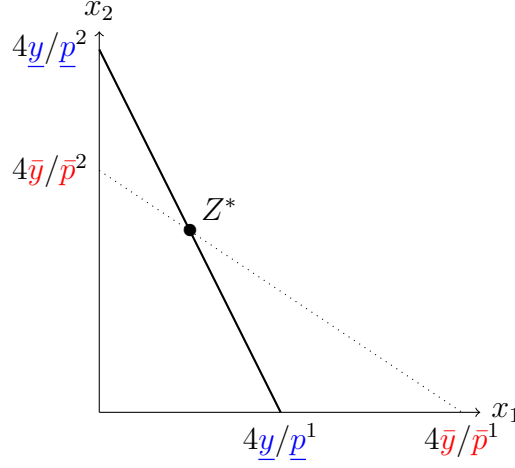
In the previous section, we constructed two economies in which all agents face the same budget within each economy, i.e. two equal-budget economies. The Laissez-faire allocations in these two economies generate the same aggregate consumption $z_1^* + z_1^* + z_3^* + z_3^* = Z^* = z_2^* + z_2^* + z_4^* + z_4^*$, yet the aggregate budget sets, depicted in Figure 2, are starkly different.

Because **Laissez-faire** induces a different ranking of allocations in these two economies, but repeated applications of **Fleming (1952) Separability** impose a stability of the ordering, we reached an impossibility. Arguably, Separability axiom is too strong in this context:

¹⁰I note that Bosmans and Öztürk (2022) studied the impossibility between a laissez-faire principle and the Pareto principle. Yet, their laissez-faire principle is much stronger than mine as it holds under any initial distribution of endowments while mine restricts it to economies with $(\omega_i, p_i) = (\omega, p)$ for all $i \in \mathcal{N}$. As such, their axiom is closer to libertarianism while the present axiom is closer to the ethics of equality of opportunity.

¹¹It is not difficult to see from the proof of Theorem 1 that the impossibilities would persist if we replace **Same-Preference Anonymity** with the transfer-based axiom among same-preference agents considered in Bosmans, Decancq, and Ooghe (2018). Such an impossibility is proven in Germain (2023) for a linear production economy.

Figure 2: Aggregate budgets in the proof of Theorem 1.



The quantity of good 1 is denoted by x_1 on the horizontal axis and the quantity of good 2 is denoted by x_2 on the vertical axis. The consumption vector $Z^* \in \mathbb{R}_+^L$ is the total quantity of resources used in each Laissez-faire allocation, i.e. $z_1^* + z_1^* + z_3^* + z_3^* = Z^* = z_2^* + z_2^* + z_4^* + z_4^*$.

the social ordering should not be completely immune to changes in indifferent agents' characteristics, because they shape what is feasible from lump-sum transfers, and in turn affect how **Laissez-faire** axioms bite in equal-budget economies. At minimum, the weakening strategy must forbid applications of Separability between equal-budget economies whose aggregate budgets are not nested.

The natural way to escape the impossibility would be to impose Separability between two generic economies e and e' only if the aggregate budget of one is nested into the other. However, this is not enough. Indeed, imagine $\mathcal{B}(e)$ is nested in $\mathcal{B}(e')$ and $\mathcal{B}(e'')$ is nested in $\mathcal{B}(e')$. By repeated applications of Separability, we will be able to compare $\mathcal{B}(e)$ and $\mathcal{B}(e'')$, yet they may be non-nested simplices.

Our weakening strategy will consist of applying Separability between equal-budget $\hat{e}$ and $\hat{e}'$ economies only if their aggregate budgets are proportional to one another, that is $\mathcal{B}(\hat{e}) = \lambda \mathcal{B}(\hat{e}')$ for some $\lambda \geq 0$. Moreover, the axiom must also tell us a robustness condition for generic economies with price dispersion. For that purpose, let me define the equal-budget economy $A(e)$ associated to e as the largest (with respect to set inclusion) equal-budget economy nested in the aggregate budget of e .

$$A(e) = \max_{\subseteq} \mathcal{B}(\{(\succsim_i, \omega, p)_{i \in \mathcal{N}}\}) \subseteq \mathcal{B}(e).$$

This definition ensures that equal-budget economy's resources are as close as possible to

those of the generic economy e .¹² We are now equipped to define the logical weakening of **Fleming (1952) Separability**.

Proportional Separability: $\forall e, e' \in \mathcal{E}$ and the partition $\mathcal{K} + \mathcal{M} = \mathcal{N}$ with $e_{\mathcal{K}} = \{(\succsim_i, \omega_i, p_i)_{i \in \mathcal{K}}\}$, $e'_{\mathcal{K}} = \{(\succsim'_i, \omega'_i, p'_i)_{i \in \mathcal{K}}\}$, $e_{\mathcal{M}} = \{(\succsim_i, \omega_i, p_i)_{i \in \mathcal{M}}\} = e'_{\mathcal{M}}$, if there exists $\lambda > 0$ such that $A(e) = \lambda A(e')$, then

$$(z_{\mathcal{K}}, z_{\mathcal{M}}) \mathbf{R}(e_{\mathcal{K}}, e_{\mathcal{M}})(z'_{\mathcal{K}}, z'_{\mathcal{M}}) \iff (z'_{\mathcal{K}}, z_{\mathcal{M}}) \mathbf{R}(e'_{\mathcal{K}}, e_{\mathcal{M}})(z'_{\mathcal{K}}, z'_{\mathcal{M}})$$

This breaks the impossibilities of Theorems 1 and 2 and the comparison of non-nested simplices depicted in Figure 2: chained applications of Separability cannot lead to comparing two equal-budget economies that are non-nested. It will be useful to understand the geometric consequences of imposing **Proportional Separability**. They are captured in the following lemma.

Lemma 1. *Any two economies e and e' are such that $A(e) = \lambda A(e')$ if and only if*

$$\text{for each } l = \{1, \dots, L\} : \sum_{i \in \mathcal{N}} \frac{y_i}{p_i^l} = \lambda \sum_{i \in \mathcal{N}} \frac{y'_i}{p_i'^l}$$

Proof. Any equal-budget economy has an aggregate budget defined by $\{Z \in \mathbb{R}_+^L : pZ \leq Ny\}$. This is a simplex whose intercept with the axis l is Ny/p^l . The intercept of the generic economy aggregate budget $\mathcal{B}(e)$ with the axis l is $\sum_{i \in \mathcal{N}} \frac{y_i}{p_i^l}$. Because $\mathcal{B}(e)$ is convex and contains all the points below its boundary, we have

$$\mathcal{B}(\{(\succsim_i, \omega, p)_{i \in \mathcal{N}}\}) \subseteq \mathcal{B}(e) \iff Ny/p^l \leq \sum_{i \in \mathcal{N}} \frac{y_i}{p_i^l} \text{ for each } l = \{1, \dots, L\}$$

The largest equal-budget economy that satisfies the set inclusion is therefore the one where the inequality is saturated. Hence, $A(e) = \lambda A(e')$ if and only if $\sum_{i \in \mathcal{N}} \frac{y_i}{p_i^l} = \lambda \sum_{i \in \mathcal{N}} \frac{y'_i}{p_i'^l}$, which is the stated condition. ■

This lemma shows that our weakening of Separability is equivalent to comparing economies that have proportional resources, if one defines resources in good l as $\sum_{i \in \mathcal{N}} \frac{y_i}{p_i^l}$, that is the maximal quantity of good l that could ever be consumed in that economy. Economically, the aggregate consumption of good l would be $\sum_{i \in \mathcal{N}} \frac{y_i}{p_i^l}$ if every agent was spending all her income in that good. Geometrically, the resource in good l is the intercept of the aggregate budget with the l -axis. I use the terminology *resources* because in our context of price dispersion, both individual prices and endowments matter for the feasibility of an allocation.

¹²A separate literature on *freedom of choice* axiomatized rankings of individual budget sets with heterogeneity in both income and prices but without preferences. For example, two budgets are equivalent if they have the same volume in Xu (2004) or if they cross a predetermined ray at the same point in Kolm (2010). None of these two propositions would escape the impossibility in our context, because they would still allow the comparison of non-nested simplices. Moreover, our axiom bears on aggregate budgets rather than individual budgets.

The main theorems of this paper will make use of **Proportional Separability** as well as **Same-Preference Anonymity**. However, they will feature two logically stronger versions of the *laissez-faire* principle, which I now define in turn. Unlike **Laissez-faire^W**, these axioms are able to rank any pair of feasible allocations in the equal-budget economies.

Egalitarian Responsibility imposes that any reduction of inequality towards the Laissez-faire allocation is a welfare improvement in equal-budget economies. Formally, at any feasible allocation where two agents consume different bundles, we compute the budgets tangent to the agent's indifference curves at these bundles and a reduction of inequality between these budgets is welfare-improving.

Egalitarian Responsibility: $\forall e \in \mathcal{E}$ such that $(\omega_i, p_i) = (\omega, p) \forall i \in \mathcal{N}$, and z, z' feasible by lump-sum transfers from e with $z_k = z'_k \forall k \in \mathcal{N} \setminus \{i, j\}$, and for $r = \{i, j\}$ we have the agent r 's indifference curve through z_r and z'_r are tangent to some $B(t_r, \omega, p)$ and $B(t'_r, \omega, p)$ respectively, if there exists $\delta \geq 0$ such that $t'_i + \delta = t_i \leq t_j = t'_j - \delta$, then $z\mathbf{R}(\mathbf{e})z'$

Similarly, **Utilitarian Responsibility** computes the budgets tangent to the agents' indifference curve at any allocation. However, a welfare-improvement arises if the sum of these budgets has increased.

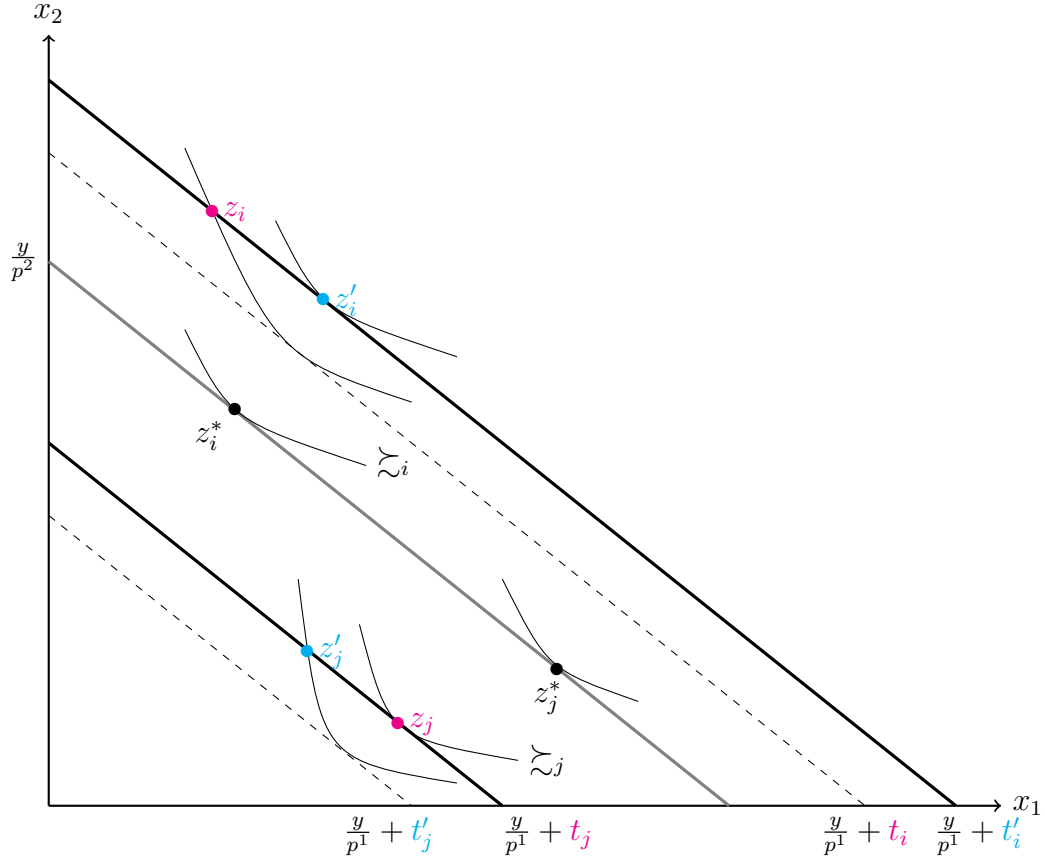
Utilitarian Responsibility: $\forall e \in \mathcal{E}$ such that $(\omega_i, p_i) = (\omega, p) \forall i \in \mathcal{N}$, and z, z' feasible by lump-sum transfers from e with $z_k = z'_k \forall k \in \mathcal{N} \setminus \{i, j\}$, and for $r = \{i, j\}$ we have the agent r 's indifference curve through z_r and z'_r are tangent to some $B(t_r, \omega, p)$ and $B(t'_r, \omega, p)$ respectively, if $t_i + t_j \geq t'_i + t'_j$, then $z\mathbf{R}(\mathbf{e})z'$

The difference between these two axioms is illustrated in Figure 3. I built a two-agent economy and two allocations z and z' feasible from lump-sum transfers such that z is obtained by an inequality reduction from z' . By construction, **Egalitarian Responsibility** imposes $z\mathbf{R}(\mathbf{e})z'$. However, this transfer between agents does not change the sum of transfers, such that **Utilitarian Responsibility** imposes $z\mathbf{I}(\mathbf{e})z'$.

It is important to note that both **Egalitarian Responsibility** and **Utilitarian Responsibility** imply that the Laissez-faire allocation z^* is optimal. Indeed, the *laissez-faire* allocation is such that there is no lump-sum transfer inequality among agents, which **Egalitarian Responsibility** finds most desirable. However, the *laissez-faire* allocation is also the one where agents choose their preferred bundle while the sum of lump-sum transfers is 0, which is preferred to all other feasible allocations (i.e. $\sum t_i \leq 0$) by **Utilitarian Responsibility**. Both axioms confirm the supremacy of *laissez-faire* allocation in equal-budget economies, but for different reasons. They also have bite beyond that as any change toward the *laissez-faire* allocation is desirable, thus being logically stronger than **Laissez-faire^W**. As such, using any of them together with **Same-Preference Anonymity**, **Weak Pareto** and **Fleming (1952) Separability** would lead us to the impossibility of Theorem 2.

I now present the main theorems of the paper. They show that using **Proportional Separability** escapes the impossibility. I start with two theorems that partially characterize

Figure 3: In this economy with equal prices and endowments, **Egalitarian Responsibility** implies $(z_i, z_j) \mathbf{R}(\mathbf{e}) (z'_i, z'_j)$ but **Utilitarian Responsibility** implies that $(z_i, z_j) \mathbf{I}(\mathbf{e}) (z'_i, z'_j)$.



Notes: This figure is constructed such that both agent i and agent j have identical prices p and income y but different preferences. The Laissez-faire budget is in gray and the solid black lines are equidistant to it. Hence, (z_i, z_j) and (z'_i, z'_j) are both feasible by a lump-sum transfer from j to i . The dashed lines are the budgets tangent to the indifference curve of the agents at the particular bundle. Because $t_j = \delta + t'_j$ and $t_i = t'_j - \delta$ **Egalitarian Responsibility** implies that $(z_i, z_j) \mathbf{R}(\mathbf{e}) (z'_i, z'_j)$. However, because such δ necessarily implies that $t'_i + t'_j = t_i + t_j$, **Utilitarian Responsibility** implies that $(z'_i, z'_j) \mathbf{I}(\mathbf{e}) (z_i, z_j)$.

how the social welfare function should look like. Both theorems pin down one possibility, and each one uses the same measure of individual consumer welfare, but they differ in their aggregation rule. The individual consumer welfare is the expenditure function with a common reference price for all consumers. Formally, it is defined as

$$W_i(\tilde{p}, z_i) = \min_{z_0} \tilde{p} \cdot z_0 \text{ such that } z_0 \succsim_i z_i \text{ and } \tilde{p}^l = \frac{\sum_{i \in \mathcal{N}} y_i}{\sum_{i \in \mathcal{N}} \frac{y_i}{p_i^l}} \text{ for } l = \{1, \dots, L\}$$

Let me stress that $W_i(\cdot)$ represents agent i 's preferences: a preferred bundle increases $W_i(\cdot)$. Moreover, it is also money-metric because it is expressed in the same unit as income. I provide a graphical illustration of the construction of $W_i(\tilde{p}, z_i)$ for a two-agent case with two goods in Figure 4. In panel (a), I show that $\tilde{p}^l$ is the price that renders the total resources of the economy affordable with the aggregate income $\sum_i y_i$. In panel (b), I show that the welfare of agent i consuming z_i is the minimal expenditure at price $\tilde{p}$ required for agent i to be indifferent to z_i .

Theorem 3 shows that **Egalitarian Responsibility**, combined with **Same-Preference Anonymity** and **Proportional Separability**, leads to a social welfare function that has inequality aversion in this money-metric welfare measure.

Theorem 3. *If $R(e)$ satisfies **Egalitarian Responsibility**, **Same-Preference Anonymity**, and **Proportional Separability**, then for z, z' feasible by lump-sum transfers from e and a $\delta \geq 0$ such that*

$$W_i(\tilde{p}, z'_i) + \delta = W_i(\tilde{p}, z_i) \leq W_j(\tilde{p}, z_j) = W_j(\tilde{p}, z'_j) - \delta$$

while $z'_k = z_k$ for all $k \in \mathcal{N} \setminus \{i, j\}$, then $z \mathbf{R}(e) z'$.

The novel constructive proof is relegated to Appendix A and its strategy is as follows. I start in an equal-budget economy with agents i and j respecting the premise and show that **Egalitarian Responsibility** yields a ranking of allocations. The goal is to extend that ranking to a generic economy.¹³

The next theorem shows the consequences of replacing **Egalitarian Responsibility** by **Utilitarian Responsibility**. Its proof is relegated to the appendix as well.

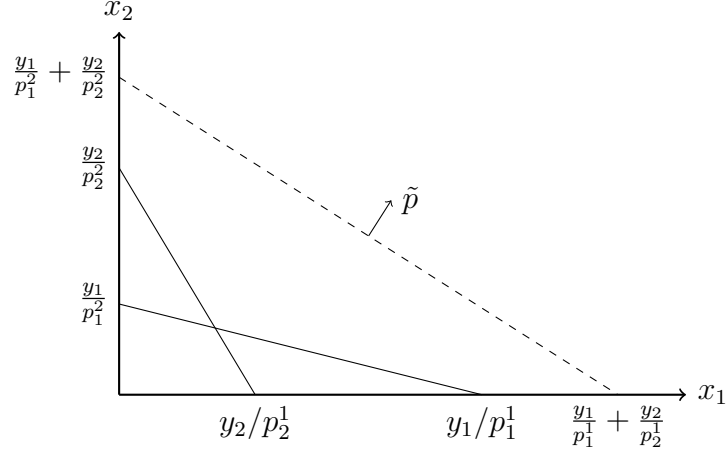
Theorem 4. *If $R(e)$ satisfies **Utilitarian Responsibility**, **Same-Preference Anonymity**, and **Proportional Separability** then for z, z' feasible by lump-sum transfers from e such that*

$$W_i(\tilde{p}, z_i) + W_j(\tilde{p}, z_j) \geq W_i(\tilde{p}, z'_i) + W_j(\tilde{p}, z'_j)$$

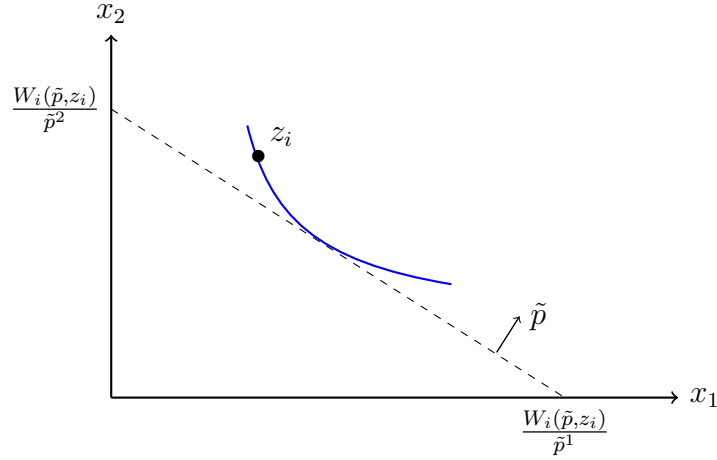
while $z_k = z'_k \forall k \in \mathcal{N} \setminus \{i, j\}$, then $z \mathbf{R}(e) z'$.

¹³If I had weaken Separability to apply only between economies with identical set of feasible allocations rather than identical aggregate budgets, there would have been no way to extend the ranking from an equal-budget economy to a generic economy, because generic economies necessarily have larger feasibility sets as discussed after Remark 1.

Figure 4: Illustrations of the construction of the interpersonal consumer welfare measure.



(a) Illustration of the construction of $\tilde{p}$.



(b) Illustration of the construction of $W_i(\tilde{p}, z_i)$.

Panel (a) shows that $\tilde{p}$ is the common price vector that makes the total quantity of resources in the economy affordable with the aggregate income $y_1 + y_2$. Panel (b) shows that the welfare $W_i(\tilde{p}, z_i)$ of agent i consuming z_i is the amount of expenditures at price $\tilde{p}$ needed to be indifferent to z_i .

Although these theorems give suggestions on the shape of the social welfare function, the reader may not find them satisfactory because they do not characterize a full social welfare function: they only bite when two agents differ, not in the general case where many agents differ. The subsequent two theorems show that we can retrieve a full characterization of the social welfare function by adding standard axioms. Their proofs, which are not novel and build on known strategies from d'Aspremont and Gevers (1977), Hammond (1976), and Milnor (1951), are also relegated to Appendix A.

Theorem 5. *$R(e)$ satisfies Egalitarian Responsibility, Same-Preference Anonymity, Proportional Separability, and Strong Pareto, if and only if for any z, z' feasible by lump-sum*

transfers from e ,

$$(W_i(\tilde{p}, z_i))_{i \in \mathcal{N}} \geq_{lex} (W_i(\tilde{p}, z'_i))_{i \in \mathcal{N}} \iff z \mathbf{R}(e) z'$$

where $\geq_{lex}$ is the minimum ordering applied lexicographically.

Even though this theorem characterizes a social welfare function with an infinite aversion to inequality in the particular money-metric utility $W_i(\tilde{p}, z_i)$, observe that the optimal allocation still displays some consumption inequalities. Indeed, the optimal allocation which $\max \min_{i \in \mathcal{N}} W_i(\tilde{p}, z_i)$ is one where everyone's indifference curves are tangent to the same budget defined by prices $\tilde{p}$ and income level $W_i(\tilde{p}, z_i^*) = W_j(\tilde{p}, z_j^*)$ for any agent $i, j \in \mathcal{N}$. Because everyone has different preferences, this tangency can be achieved even if bundles are unequal. This shows that these axioms reflect the ethics of equality of *opportunity* and not of *outcomes*: budget sets should be equalized but not consumption vectors.

The next result completely characterizes utilitarianism with our axiomatic theory.

Theorem 6. *$\mathbf{R}(e)$ satisfies **Utilitarian Responsibility**, **Same-Preference Anonymity**, **Proportional Separability**, and **Strong Pareto** if and only if for any z, z' feasible by lump-sum transfers from e ,*

$$\sum_{i \in \mathcal{N}} W_i(\tilde{p}, z_i) \geq \sum_{i \in \mathcal{N}} W_i(\tilde{p}, z'_i) \iff z \mathbf{R}(e) z'$$

These results deserve some comments. First, these theorems prove that **Proportional Separability** allowed us to escape the impossibilities raised in Theorem 1 and 2. These possibilities differ in the aggregator used to compute social welfare (the *maximin* and the *sum*, respectively), but they use the very same measure of interpersonal welfare, i.e. $W_i(\tilde{p}, z_i)$. It suggests that this money-metric utility function should be used to compare the welfare of heterogeneous consumers if one abides by **Same-Preference Anonymity**, **Laissez-faire**^W and **Proportional Separability**, while the choice of the aggregator depends on which logical strengthening of the Laissez-faire principle the ethical observer prefers between **Utilitarian Responsibility** and **Egalitarian Responsibility**. Whether the observer should use the sum or the maximin to assess collective welfare changes in any economy can be reduced to the problem of evaluating collective welfare between equally-endowed individuals. If the planner wants to minimize welfare inequalities among same-budget individuals, then the maximin should be adopted as social welfare aggregator. If the observer rather wishes to maximize the total welfare of same-budget individuals, then it should use the summation.

Second, these possibilities only uses ordinal and non-comparable elements of preferences to build a welfare measure. Indeed, the only component of preferences needed to compute $W_i(\tilde{p}, z_i)$ is the indifference curve at z_i , not the utility level. Despite that, this measure allows interpersonal comparisons of welfare between individuals. Thus, this measure is in line with the long-standing tradition since Robbins (1938) and Samuelson (1947) of not comparing cardinal utility levels across individuals.¹⁴

¹⁴This is not a new result in itself, as it is characteristic of the fairness literature in social choice. See Bosmans, Decancq, and Ooghe (2018), Fleurbaey (2003), Fleurbaey and Maniquet (2011), and Piacquadio (2017) among others.

Third, that the sum of expenditure functions is an attractive way of measuring welfare has a long history in welfare economics marked with controversies over the choice of the reference price vector. First, the Hicksian compensating and equivalent variations are expenditure functions using post- and pre-reform prices as reference situation respectively¹⁵, but a social welfare function using their sum may lead to intransitive assessment as is known since Boadway (1974) and Scitovsky (1941).¹⁶ Second, money-metric utility functions have been criticized by Blackorby and Donaldson (1988) because they may yield anti-redistributive consequences for some reference prices¹⁷ even when the social welfare function is egalitarian. Third, Blackorby, Laisney, and Schmachtenberg (1993) and Roberts (1980) underlined the arbitrariness of the choice of the reference price and studied price-independent welfare assessment. What the present paper offers are social welfare functions based on money-metric utility that overcome each of these critiques: they are transitive, allow for inequality aversion, and their reference situation is endogenized by the axioms. Moreover, $\tilde{p}$ is independent of the normalization of the numéraire of the economy. I now present some consequences for applied welfare analysis.

5 Applications

The previous results showed that (i) collective welfare aggregates individual welfare with a sum or a maximin, (ii) the individual welfare measure compare individual with heterogeneous preferences using an expenditure function $W_i(\cdot)$, and (iii) the expenditure function deflates each countries' equivalent consumption by the same reference price $\tilde{p}$ which the income-weighted harmonic mean of individual prices.

In terms of practical applications, the choice of the aggregator is ultimately a normative choice. Moreover, the present paper does not improve on existing empirical techniques to recover expenditures functions and money-metric utilities as do recent contributions by Atkin et al. (2024), Baqaee, Burstein, and Koike-Mori (2024), Gauthier, Maes, and Malhotra (2023), and Jaravel and Lashkari (2024). However, the determination of $\tilde{p}$ is new and I show that it can help in a range of problems to compare consumers interpersonally. For the remainder of the paper, I abstract from collective welfare and focus on interpersonal comparison of welfare.

¹⁵The widely-used consumer surplus is a Hicksian variation when preferences are quasilinear. See Hicks (1939, 1941, 1942, 1943, 1945) for the original contributions.

¹⁶See Blackorby and Donaldson (1990) for a review. Recently, Weretka and Dec (2026) showed that when preferences are von Neumann-Morgenstern in macro models, this problem increases with the degree of consumers' impatience.

¹⁷Bosmans, Decancq, and Ooghe (2018), Fleurbaey and Maniquet (2011), and Schlee and Khan (2022, 2023) discussed several ways to escape this critique.

5.1 Income of nations

In the index number problem¹⁸ initiated by Fisher (1922), the economist must compare the *real income* of a set of countries based on a dataset of observed prices p_i and quantities consumed z_i and decompose this real income into a price index and a quantity index. Our analysis of welfare above can readily be applied to this problem if each country is considered as an agent and *real income* is considered as the expenditure function $W_i(\tilde{p}, z_i)$. More formally, real income can be decomposed into a price index P_i and a quantity index $Q_{i,j}$ as follows

$$P_i = \frac{W_i(p_i, z_i)}{W_i(\tilde{p}, z_i)}$$

$$Q_{i,j} = \frac{W_i(\tilde{p}, z_i)}{W_j(\tilde{p}, z_j)}$$

where the price index P_i reflects the deflation needed to compare the prices faced by country i and $Q_{i,j}$ captures the real income difference between countries i and j .¹⁹

The present paper argues for a common price $\tilde{p}$ to deflate countries' consumption. Interestingly, the widely-used Geary (1958) method, which underpins the Penn World Table and the international dollar, also computes a world price²⁰ $\hat{p}^l$ for each good l defined as

$$\hat{p}^l = \frac{\sum_i p_i^l z_i^l \epsilon_i}{\sum_i z_i^l}$$

where ϵ_i is a normalization of countries' income such that $\epsilon_i = \frac{\sum_l \hat{p}^l z_i^l}{\sum_l p_i^l z_i^l}$. The real income of country i is then computed as the cost of that country's bundle z_i at the world prices $\hat{p}$. While this approach ignores preferences, Neary (2004) shows that consistency with a representative agent preferences can be obtained by setting ϵ_i as the ratio of expenditure functions at these prices.

A necessary condition for $\tilde{p}$ to be identical to $\hat{p}$ is that, for each good l , expenditure shares $\frac{p_i^l z_i^l}{y_i}$ are identical across countries. This is quite unlikely to hold in any international comparisons: we know since Engel (1857) that richer countries spend proportionally less on food. As a consequence, we should expect our modified PPP to yield different empirical results than the conventional Geary method.

Moreover, observe that if there is a preference shock to one country while all prices and income in the world are constant, Geary's prices $\hat{p}^l$ will change, while this paper's $\tilde{p}$ will

¹⁸That expenditure functions and welfare economics are useful for index number theory is known at least since the classical contributions of Deaton (1979) and Samuelson and Swamy (1974).

¹⁹In the literature, the former is a true Köns price index while the latter is an Allen true quantity index. This paper's measure escapes the Van Veelen (2002) impossibility by using third-country information when computing the real income of a pair of countries.

²⁰Deaton and Heston (2010) argued that the main advantage of the approach is the preservation of integrability. Indeed, because it uses a single world price for each good, a summation across goods does not perturb the ranking.

not. Indeed, the former depends on expenditure shares of each country, while the latter only depends on prices and income. Because of this, the deflation proposed in this paper is robust to preference heterogeneity. Importantly, this modification of the purchasing power parity does not require more data than what is commonly used.

I note that the computation of $W_i(\tilde{p}, z_i)$ requires to know agent i 's indifference curve through z_i . Many applications only care about prices and quantities abstracting from indifference curves, be that for normative reasons or data constraints. This is equivalent to assuming that preferences are Leontief. In that case, the expenditure function at z_i is exactly the total amount of money spend at z_i .

5.2 Lifetime inflation and growth

We can also interpret the standard consumption space over time by considering each good l is a time period. For clarity, let me substitute the superscript l by $t = \{1, \dots, T\}$. The problem for the growth economist consists in measuring inflation and real income growth over the past T periods for a set of countries based on a dataset of country i 's prices p_i^t and consumption z_i^t . Preferences are defined over the T periods, i.e. over the stream of consumption z_i , and must be understood as *lifetime* welfare.

The present paper suggests that the welfare-relevant inflation and growth indices between consumers i and j over the T periods are

$$\Pi_i = \frac{W_i(p_i, z_i)}{W_i(\tilde{p}, z_i)}$$

$$Q_{i,j} = \frac{W_i(\tilde{p}, z_i)}{W_j(\tilde{p}, z_j)}$$

Interestingly, in such case the reference price vector $\tilde{p}$ becomes close to the GDP deflator i.e. $\tilde{p}^t = \frac{\sum_i y_i}{\sum_i y_i / p_i^t}$.²¹ Unlike the Consumer Price Index (CPI), the GDP deflator does not depend on the weights of each consumption good in the bundle consumed, but rather only on income and prices. Again, this can be directly traced back to the axioms: **Laissez-faire** precludes the economist from choosing a particular consumption good like the CPI basket, because it would give prominence to a particular kind of preferences while all preferences should be deemed equal.

Importantly, this does not mean that preferences are ignored. Indeed, Π_i and $Q_{i,j}$ both depend on preferences as they use expenditure functions as arguments. However, the measuring rod does not: $\tilde{p}$ only depends on consumption possibilities.

²¹In practice, the GDP deflator is computed on more goods than mere private consumption goods and include the price of investments as well as government expenditures.

5.3 Year-by-year inflation and growth

Consider the case where each time period the preferences of the agent under study may vary. In such case, we can consider that the index i used above may be substituted by t . In turn, measuring welfare interpersonally becomes a measure of year-by-year inflation and welfare growth for a single country when prices, endowments, and preferences may change from one year to the next. The relevant inflation and growth indices are

$$\Pi_t = \frac{W_t(\tilde{p}, z_t)}{W_t(p_t, z_t)}$$

$$Q_{t,t-1} = \frac{W_t(\tilde{p}, z_t)}{W_{t-1}(\tilde{p}, z_{t-1})}$$

In this setup, the reference price for a good l $\tilde{p}^l$ becomes the ratio between the sum of expenditures over time over the the sum of maximal quantity of good l that could have been consumed.

$$\tilde{p}^l = \frac{\sum_t y_t}{\sum_t y_t / p_t^l}$$

While traditional inflation and growth measures typically ignore preferences or assumed homothetic preferences, a recent contribution by Jaravel and Lashkari (2024) provides an algorithm to measure inflation and growth for non-homothetic and general preferences. Interestingly, they also rely on expenditure functions with a common, time-invariant reference price vector which they call *base period*. However, neither their theory nor their algorithm provides guidance on which base period to choose. While their estimates significantly differ from standard estimates derived under homothetic preferences, the magnitude of these differences depends on the *base period*. What the present paper gives is a normative justification for the base period that connects the measure with consumer welfare.

6 Conclusion

This paper showed that, once heterogeneous consumers face different prices, interpersonal comparison, deflation, and aggregation can be jointly determined. If consumption inequalities due to heterogeneous preferences do not hurt social welfare, same-preference consumers are treated anonymously, and social welfare is consistent across economies, then individual consumer welfare is evaluated by the expenditure required to reach her indifference curve at a common price vector. We showed that this common price vector is the income-weighted harmonic mean of individual prices, and it can be retrieved without additional data in various applications.

The broader goal of the paper was to show that statistics on the costs and standards of living embed normative choices about how heterogeneous consumers should be compared. By linking the current practice of statistical offices with welfare economics, the paper hopes to make these choices more explicit.

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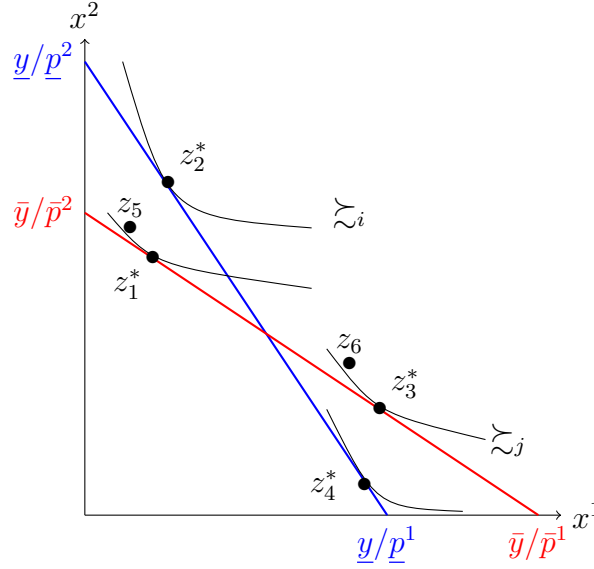
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A Proofs

Proof of Theorem 2. I proceed again by contradiction. Consider the same setup as in the proof of Theorem 1 and z_5, z_6 such that (z_5, z_5, z_6, z_6) is feasible by lump-sum transfers from the economy where all agents have $B(0, \underline{\omega}, \underline{p})$ while $z_5 \succ_i z_1^*$ while $z_6 \succ_j z_3^*$. This is illustrated in Figure A.1.

Figure A.1: Illustration of the proof of Theorem 2.



Notes: The quantity of good 1 is denoted by x_1 on the horizontal axis and the quantity of good 2 is denoted by x_2 on the vertical axis. The economies have been chosen such that the Laissez-faire allocation in the blue economy is feasible by lump sum transfers from the red economy, and vice versa, that is $\frac{z_1 + z_3}{2} = \frac{z_2 + z_4}{2}$. Moreover, (z_5, z_5, z_6, z_6) is feasible by lump sum transfers from the when all agents have $B(0, \underline{\omega}, \underline{p})$.

Using the exact same steps as in Theorem 1 but with a weak preference relation rather than a strict one, we get that by *Laissez-Faire^W*, *Same-Preference Anonymity*, Transitivity and **Fleming (1952) Separability**,

$$(z_2^*, z_1^*, z_3^*, z_4^*) \quad \mathbf{R} \left(\left\{ (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}) \right\} \right) \quad (z_2^*, z_2^*, z_4^*, z_4^*) \quad (\text{A.1})$$

By **Fleming (1952) Separability**,

$$(z_1^*, z_1^*, z_3^*, z_3^*) \quad \mathbf{R} \left(\left\{ (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}) \right\} \right) \quad (z_1^*, z_2^*, z_4^*, z_3^*)$$

By **Same-Preference Anonymity** and Transitivity,

$$(z_1^*, z_1^*, z_3^*, z_3^*) \quad \mathbf{R} \left(\left\{ (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}) \right\} \right) \quad (z_2^*, z_1^*, z_3^*, z_4^*)$$

By Transitivity with equation A.1,

$$(z_1^*, z_1^*, z_3^*, z_3^*) \quad \mathbf{R} \left(\left\{ (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}) \right\} \right) \quad (z_2^*, z_2^*, z_4^*, z_4^*)$$

By **Laissez-Faire**^W,

$$(z_2^*, z_2^*, z_4^*, z_4^*) \quad \mathbf{R} \left(\left\{ (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}) \right\} \right) \quad (z_5, z_5, z_6, z_6)$$

By Transitivity,

$$(z_1^*, z_1^*, z_3^*, z_3^*) \quad \mathbf{R} \left(\left\{ (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_i, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}), (\succsim_j, \underline{\omega}, \underline{p}) \right\} \right) \quad (z_5, z_5, z_6, z_6)$$

which contradicts **Weak Pareto** and completes the proof. ■

Proof of Theorem 3. The proof proceeds by construction. Take a generic economy $e \in \mathcal{E}$ and two allocations z, z' feasible lump-sum transfers such that the premise holds. Define the equal-budget economy $\{(\succsim_i, \tilde{\omega}, \tilde{p})_{i \in \mathcal{N}}\}$ where $\tilde{p}^i = \frac{\sum_{i \in \mathcal{N}} y_i}{\sum_{i \in \mathcal{N}} \frac{y_i}{p_i^i}}$ and $\tilde{\omega}$ large enough such that z and z' are feasible from lump-sum transfers in that equal-budget economy. Take $t_r = W_r(\tilde{p}, z_r) - \tilde{y}$ and $t'_r = W_r(\tilde{p}, z'_r) - \tilde{y}$ for $r = \{i, j\}$. Because of the premise $W_i(\tilde{p}, z'_i) + \delta = W_i(\tilde{p}, z_i) \leq W_j(\tilde{p}, z_j) = W_j(\tilde{p}, z'_j) - \delta$, we have $t'_i + \delta = t_i \leq t_j = t'_j - \delta$. Without loss of generality, I relabel agents $i = 3$ and $j = 4$. By **Egalitarian Responsibility**, we have

$$z \quad \mathbf{R} \left(\left\{ (\succsim_1, \tilde{\omega}, \tilde{p}), (\succsim_2, \tilde{\omega}, \tilde{p}), (\succsim_3, \tilde{\omega}, \tilde{p}), (\succsim_4, \tilde{\omega}, \tilde{p}), (\succsim_i, \tilde{\omega}, \tilde{p})_{i \in \{5, \dots, N\}} \right\} \right) (z_1, z_2, z'_3, z'_4, \dots, z_N)$$

By **Proportional Separability** for $\lambda = 1$, changing the preferences of the first two indifferent agents preserves the ordering,

$$z \quad \mathbf{R} \left(\left\{ (\succsim_3, \tilde{\omega}, \tilde{p}), (\succsim_4, \tilde{\omega}, \tilde{p}), (\succsim_3, \tilde{\omega}, \tilde{p}), (\succsim_4, \tilde{\omega}, \tilde{p}), (\succsim_i, \tilde{\omega}, \tilde{p})_{i \in \{5, \dots, N\}} \right\} \right) (z_1, z_2, z'_3, z'_4, \dots, z_N)$$

By **Same-Preference Anonymity** applied twice, permuting bundles between the first and third agents, as well as between the second and fourth agents, leaves society indifferent,

$$(z'_3, z'_4, z_1, z_2, \dots, z_N) \mathbf{I} \left(\left\{ (\succsim_3, \tilde{\omega}, \tilde{p}), (\succsim_4, \tilde{\omega}, \tilde{p}), (\succsim_3, \tilde{\omega}, \tilde{p}), (\succsim_4, \tilde{\omega}, \tilde{p}), (\succsim_i, \tilde{\omega}, \tilde{p})_{i \in \{5, \dots, N\}} \right\} \right) (z_1, z_2, z'_3, z'_4, \dots, z_N)$$

By transitivity,

$$z \quad \mathbf{R} \left(\left\{ (\succsim_3, \tilde{\omega}, \tilde{p}), (\succsim_4, \tilde{\omega}, \tilde{p}), (\succsim_3, \tilde{\omega}, \tilde{p}), (\succsim_4, \tilde{\omega}, \tilde{p}), (\succsim_i, \tilde{\omega}, \tilde{p})_{i \in \{5, \dots, N\}} \right\} \right) (z'_3, z'_4, z_1, z_2, \dots, z_N)$$

By **Same-Preference Anonymity** applied twice,

$$z \quad \mathbf{I} \left(\left\{ (\succsim_3, \tilde{\omega}, \tilde{p}), (\succsim_4, \tilde{\omega}, \tilde{p}), (\succsim_3, \tilde{\omega}, \tilde{p}), (\succsim_4, \tilde{\omega}, \tilde{p}), (\succsim_i, \tilde{\omega}, \tilde{p})_{i \in \{5, \dots, N\}} \right\} \right) (z_3, z_4, z_1, z_2, \dots, z_N)$$

By transitivity,

$$(z_3, z_4, z_1, z_2 \dots z_N) \mathbf{R} \left(\left\{ (\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_i, \tilde{\omega}, \tilde{p})_{i \in \{5, \dots, N\}} \right\} \right) (z'_3, z'_4, z_1, z_2 \dots z_N)$$

By **Proportional Separability**,

$$(z_3, z_4, z_1, z_2 \dots, z_N) \mathbf{R} \left(\left\{ (\succsim_3, \hat{\omega}, \hat{p})(\succsim_4, \hat{\omega}, \hat{p})(\succsim_k, \omega_k, p_k)_{k \in \{3, 4\}} (\succsim_i, \omega_i, p_i)_{i \in \{5, \dots, N\}} \right\} \right) (z'_3, z'_4, z_1, z_2 \dots, z_N)$$

if there exists a $\lambda_1 > 0$ and $\hat{\omega}, \hat{p}$ such that

$$\text{for each } l = \{1, \dots, L\} : N \frac{\tilde{y}}{\tilde{p}^l} = \lambda_1 \left(2 \frac{\hat{y}}{\hat{p}^l} + \sum_{i \in N \setminus \{1, 2\}} \frac{y_i}{p_i^l} \right) \quad (\text{A.2})$$

By **Same-Preference Anonymity** applied four times and transitivity,

$$z \mathbf{R} \left(\left\{ (\succsim_3, \hat{\omega}, \hat{p})(\succsim_4, \hat{\omega}, \hat{p})(\succsim_k, \omega_k, p_k)_{k \in \{3, 4\}} (\succsim_i, \omega_i, p_i)_{i \in \{5, \dots, N\}} \right\} \right) (z_1, z_2, z'_3, z'_4 \dots z_N)$$

By **Proportional Separability**,

$$z \mathbf{R}(\mathbf{e})(z_1, z_2, z'_3, z'_4 \dots, z_N)$$

only if there exists a $\lambda_2 > 0$ such that

$$\text{for each } l = \{1, \dots, L\} : 2 \frac{\hat{y}}{\hat{p}^l} + \sum_{i \in N \setminus \{1, 2\}} \frac{y_i}{p_i^l} = \lambda_2 \sum_{i \in N} \frac{y_i}{p_i^l} \quad (\text{A.3})$$

and equations (A.2) and (A.3) are jointly satisfied if

$$\text{for each } l = \{1, \dots, L\} : N \frac{\tilde{y}}{\tilde{p}^l} = (\lambda_1 * \lambda_2) \sum_{i \in N} \frac{y_i}{p_i^l}$$

which holds by construction of $\tilde{p}$ and Lemma 1 after choosing appropriate λ_1 and λ_2 . Finally, existence of $\hat{\omega}$ and $\hat{p}$ defined by (A.3) is guaranteed by setting our last degree of freedom λ_2 such that $\hat{y}/\hat{p} > 0$.

$$\lambda_2 > \max_{l \in \{1, \dots, L\}} \frac{\sum_{i \in N \setminus \{1, 2\}} y_i / p_i^l}{\sum_{i \in N} y_i / p_i^l}$$

Because $(z_1, z_2, z'_3, z'_4 \dots, z_N) = z'$, I proved $z \mathbf{R}(\mathbf{e}) z'$ and this completes the proof of Theorem 3. ■

Proof of Theorem 4. This proof is very close to the proof of Theorem 3. Take a generic economy $e \in \mathcal{E}$ and two allocations z, z' such that the premise of Theorem 4 holds. Consider the equal-budget economy $\{(\succsim_i, \tilde{\omega}, \tilde{p})_{i \in N}\}$ where $\tilde{p}^l = \frac{\sum_{i \in N} y_i}{\sum_{i \in N} \frac{y_i}{p_i^l}}$ and $\tilde{\omega}$ large enough such that z and z' are feasible from lump-sum transfers in that equal-budget economy. Let me

define $t_r = W_r(\tilde{p}, z_r) - \tilde{y}$ and $t'_r = W_r(\tilde{p}, z'_r) - \tilde{y}$ for $r = \{i, j\}$. Because of the premise $W_i(\tilde{p}, z_i) + W_j(\tilde{p}, z_j) \geq W_i(\tilde{p}, z'_i) + W_j(\tilde{p}, z'_j)$, we have $t_i + t_j \geq t'_i + t'_j$. Without loss of generality, I relabel agents $i = 3$ and $j = 4$. By **Utilitarian Responsibility**, we have

$$z \mathbf{R} \left(\left\{ (\succsim_1, \tilde{\omega}, \tilde{p})(\succsim_2, \tilde{\omega}, \tilde{p})(\succsim_3, \tilde{\omega}, \tilde{p})(\succsim_4, \tilde{\omega}, \tilde{p})(\succsim_i, \tilde{\omega}, \tilde{p})_{i \in \{5, \dots, N\}} \right\} (z_1, z_2, z'_3, z'_4, \dots, z_N) \right)$$

The remaining steps of the proof follow exactly the proof of Theorem 3. ■

Proof of Theorem 5. For necessity, if two allocations z and z' are such that $W_i(\tilde{p}, z_i) = W_i(\tilde{p}, z'_i) \forall i \in N$, then $z_i \sim_i z'_i$ for all $i \in N$ and by **Strong Pareto** we have $z \mathbf{I}(\mathbf{e}) z'$. Hence, $\mathbf{R}(\mathbf{e})$ induces an ordering defined over welfare levels $W_i(\tilde{p}, z_i)$. By Theorem 3, **Egalitarian Responsibility**, **Same-Preference Anonymity** and **Proportional Separability** imply that $\mathbf{R}(\mathbf{e})$ satisfies Hammond (1976) Equity defined over these welfare levels $(W_i(\tilde{p}, z_i))_{i \in N}$. The conclusion then follows from Hammond (1976): the induced ordering on welfare levels satisfies Hammond Equity and **Strong Pareto**, hence it coincides with the leximin ordering. For sufficiency, **Strong Pareto** is immediate. **Same-Preference Anonymity** follows because a permutation of bundles between agents with identical preferences leaves the leximin ordering of welfare unaffected, as $\succsim_i = \succsim_j$ implies $W_i(\tilde{p}, \cdot) = W_j(\tilde{p}, \cdot)$.

For **Proportional Separability**, consider $e, e' \in E$ satisfying its premises. By Lemma 1, $A(e) = \lambda A(e')$ implies that $\tilde{p} = \mu \tilde{p}'$ for some $\mu > 0$. Because expenditure functions are homogeneous of degree 1, we have $W_i(\mu \tilde{p}, \cdot) = \mu W_i(\tilde{p}, \cdot)$. Now, in the comparison between (z_K, z_M) and (z'_K, z'_M) , the welfare coordinates of agents in K are the same on both sides. Likewise, in the comparison between (z'_K, z'_M) and (z_K, z_M) , the welfare coordinates of agents in K are the same on both sides, while the welfare coordinates of agents in M are exactly μ times those in the first comparison. Since the leximin ordering is unchanged when the same list of coordinates is added to both vectors, and is also unchanged under multiplication of all coordinates by a positive scalar, the two comparisons have the same sign. Thus **Proportional Separability** holds.

Finally, consider an equal-budget economy, so that $(\omega_i, p_i) = (\omega, p)$ for all $i \in N$. Let z, z' satisfy the premises of **Egalitarian Responsibility** such that for $r \in \{i, j\}$ we have the budgets $B(t_r, \omega, p)$ and $B(t'_r, \omega, p)$ tangent to the indifference curves through z_r and z'_r , respectively. Because $\tilde{p} = p$, these tangent budgets are precisely the supporting budgets at the corresponding utility levels, so

$$W_r(\tilde{p}, z_r) = p \cdot z_r = y + t_r \quad \text{and} \quad W_r(\tilde{p}, z'_r) = p \cdot z'_r = y + t'_r$$

where $y = p \cdot \omega$. Therefore, the premise

$$t'_i + \delta = t_i \leq t_j = t'_j - \delta$$

is equivalent to

$$W_i(\tilde{p}, z'_i) \leq W_i(\tilde{p}, z_i) \leq W_j(\tilde{p}, z_j) \leq W_j(\tilde{p}, z'_j),$$

while $W_k(\tilde{p}, z_k) = W_k(\tilde{p}, z'_k)$ for all $k \in N \setminus \{i, j\}$. Hence $(W_i(\tilde{p}, z_i))_{i \in N} \geq_{\text{lex}} (W_i(\tilde{p}, z'_i))_{i \in N}$, so $z \mathbf{R}(e) z'$. Thus **Egalitarian Responsibility** holds and this completes the proof of the theorem. ■

Proof of Theorem 6. For necessity, consider a generic z and build an allocation that is identical to z except for the first two agents: z_2 is replaced by 0 and z_1 is replaced by any bundle such that agent 1's money-metric welfare becomes $W_1(\tilde{p}, z_1) + W_2(\tilde{p}, z_2)$. Such a bundle exists by continuity and local non-satiation of preferences, and $W_2(\tilde{p}, 0) = 0$. By Theorem 4, this replacement leaves social welfare unchanged. Repeating the same argument successively for agents $3, \dots, N$, it follows that z is socially indifferent to an allocation in which agents $2, \dots, N$ consume 0 and agent 1 consumes a bundle with money-metric welfare $\sum_{i \in N} W_i(\tilde{p}, z_i)$. The same argument applies to z' .

If $\sum_{i \in N} W_i(\tilde{p}, z_i) = \sum_{i \in N} W_i(\tilde{p}, z'_i)$, then z and z' are socially indifferent to two allocations in which agents $2, \dots, N$ consume 0 and agent 1 has the same money-metric welfare. By Strong Pareto which implies Pareto Indifference, these two allocations are socially indifferent, hence $zI(e)z'$.

If instead $\sum_{i \in N} W_i(\tilde{p}, z_i) > \sum_{i \in N} W_i(\tilde{p}, z'_i)$, then z and z' are socially indifferent to two allocations in which agents $2, \dots, N$ consume 0 and agent 1 has, respectively, two bundles such that the first has a strictly larger money-metric welfare than the second. Since $W_1(\tilde{p}, \cdot)$ represents $\succeq_1$, agent 1 strictly prefers the first bundle, while all other agents are indifferent. By Strong Pareto, the first allocation is socially strictly preferred to the second. Hence $zP(e)z'$. This proves that the axioms imply $z\mathbf{R}(\mathbf{e})z' \iff \sum_{i \in N} W_i(\tilde{p}, z_i) \geq \sum_{i \in N} W_i(\tilde{p}, z'_i)$.

For sufficiency, when $z\mathbf{R}(\mathbf{e})z' \iff \sum_{i \in N} W_i(\tilde{p}, z_i) \geq \sum_{i \in N} W_i(\tilde{p}, z'_i)$, **Strong Pareto** and **Same-Preference Anonymity** are obtained as in the proof of Theorem 5. For **Proportional Separability**, since the utilitarian ordering is unchanged when the same list of coordinates is added to both vectors, and is also unchanged under multiplication of all coordinates by a positive scalar, the argument follows the proof of Theorem 5 *mutatis mutandis*.

Finally, for **Utilitarian Responsibility**, consider an equal-budget economy where the premise of the axiom holds. Since $z_k = z'_k$ for all $k \in N \setminus \{i, j\}$, we have

$$\sum_{r \in N} W_r(\tilde{p}, z_r) - \sum_{r \in N} W_r(\tilde{p}, z'_r) = W_i(\tilde{p}, z_i) + W_j(\tilde{p}, z_j) - W_i(\tilde{p}, z'_i) - W_j(\tilde{p}, z'_j)$$

Because $t_i + t_j \geq t'_i + t'_j$ we have $W_i(\tilde{p}, z_i) + W_j(\tilde{p}, z_j) - W_i(\tilde{p}, z'_i) - W_j(\tilde{p}, z'_j) \geq 0$. Hence, $\sum_{i \in N} W_i(\tilde{p}, z_i) \geq \sum_{i \in N} W_i(\tilde{p}, z'_i)$, so $zR(e)z'$, proving the sufficiency and completing the proof of the theorem. $\blacksquare$